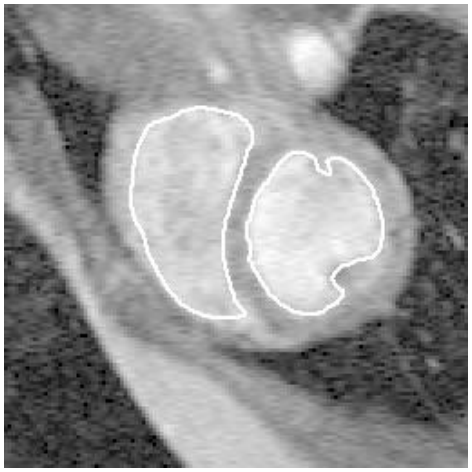
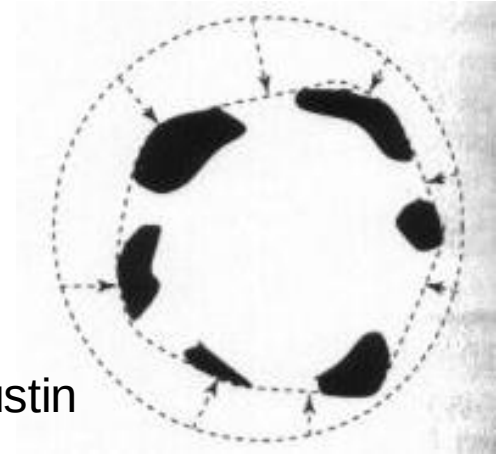


Deformable contours

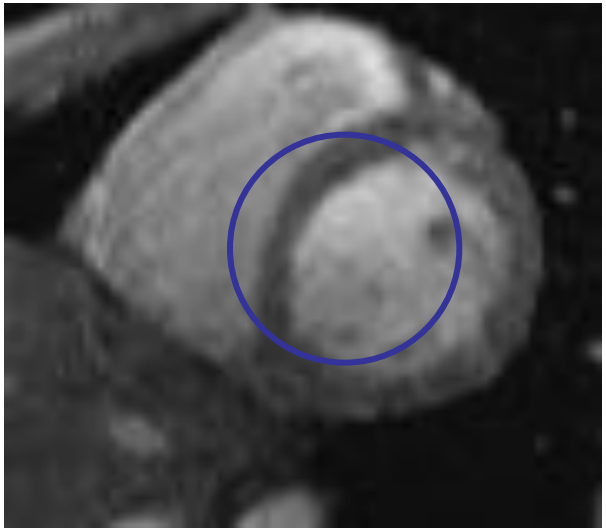


Adapted from slides by Kristen Grauman, UT Austin

Deformable contours

a.k.a. active contours, snakes

Given: initial contour (model) near desired object



(Single frame)

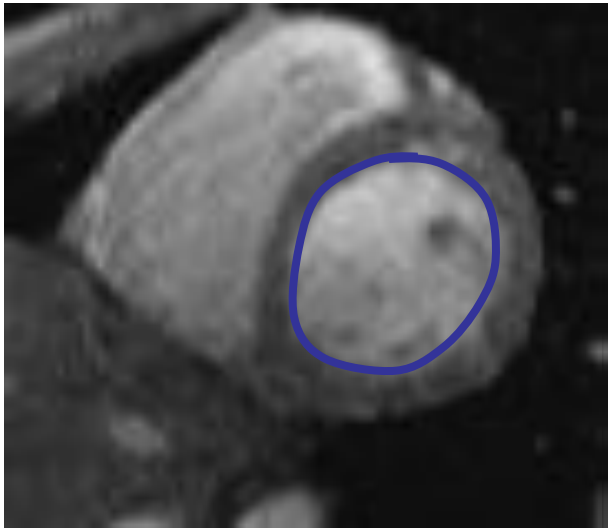
Fig: Y. Boykov

Deformable contours

a.k.a. active contours, snakes

Given: initial contour (model) near desired object

Goal: evolve the contour to fit exact object boundary



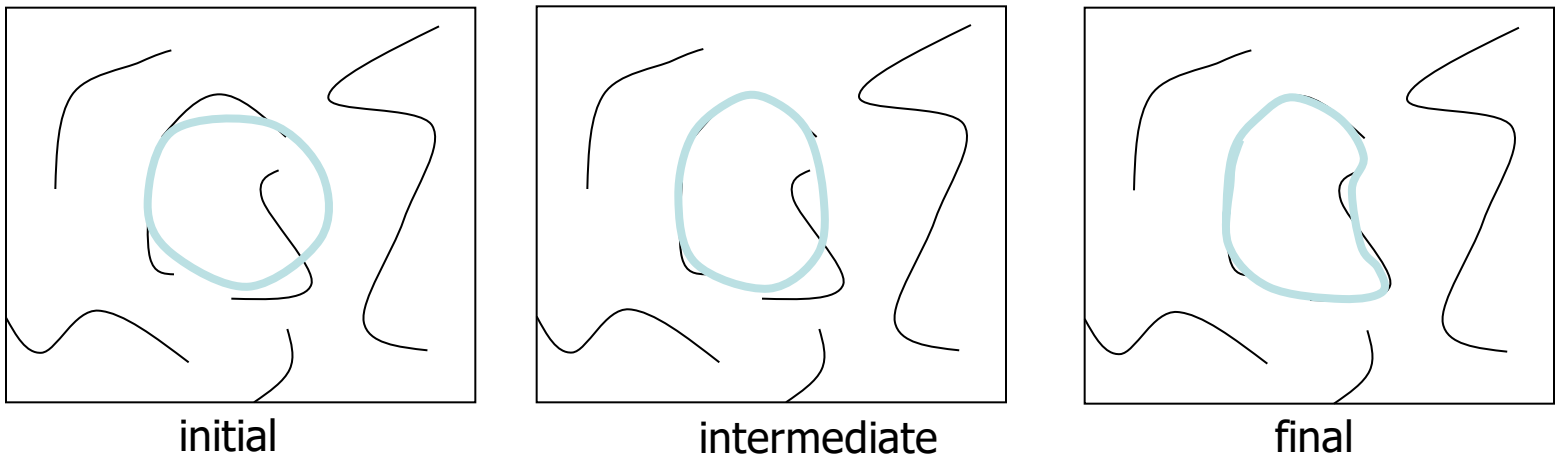
(Single frame)

[Snakes: Active contour models, Kass, Witkin, & Terzopoulos, ICCV1987]

Fig: Y. Boykov

Deformable contours

a.k.a. active contours, snakes



Initialize near contour of interest

Iteratively refine: elastic band is adjusted so as to

- be near image positions with high gradients, and
- satisfy shape “preferences” or contour priors

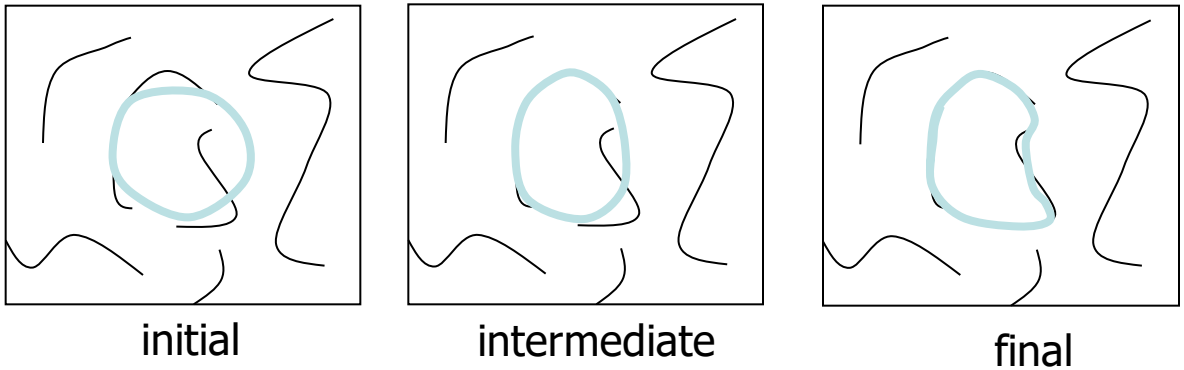
Fig: Y. Boykov

Deformable contours

a.k.a. active contours, snakes

How is the current contour adjusted to find the new contour at each iteration?

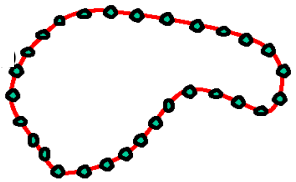
- Define a cost function (“energy” function) that says how good a possible configuration is.
- Seek next configuration that minimizes that cost function.



What are examples of problems with energy functions that we have seen previously?

Snakes energy function

The total energy (cost) of the current snake is defined as:



$$E_{total} = E_{internal} + E_{external}$$

Internal energy: encourage prior shape preferences: e.g., smoothness, elasticity, particular known shape.

External energy ("image" energy): encourage contour to fit on places where image structures exist, e.g., edges.

A good fit between the current deformable contour and the target shape in the image will yield a **low** value for this cost function.

Parametric curve representation
(continuous case)

$$v(s) = (x(s), y(s)) \qquad 0 \le s \le 1$$

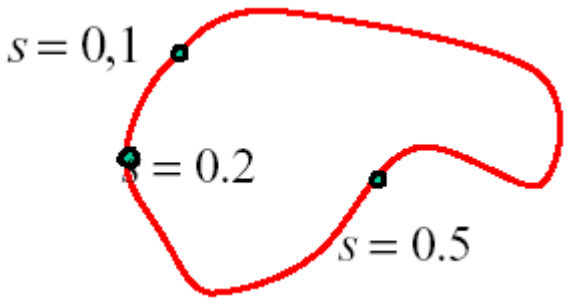
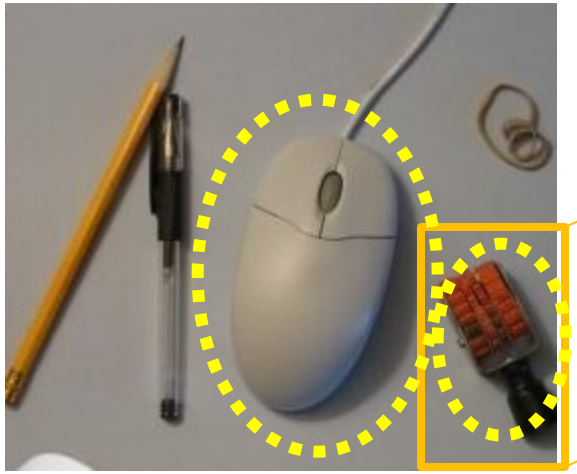


Fig from Y. Boykov

External energy: intuition

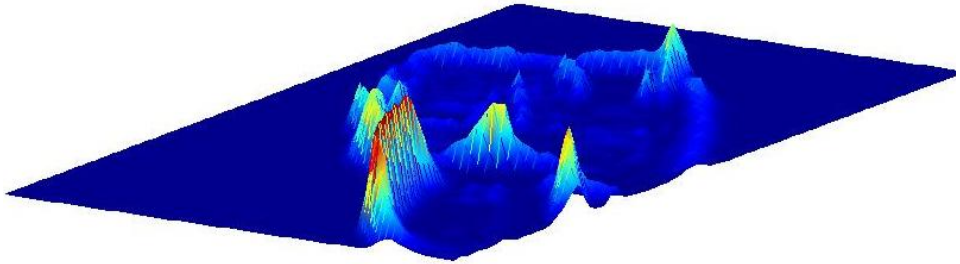
- Measure how well the curve matches the image data
- “Attract” the curve toward different image features
  - Edges, lines, etc.

External image energy



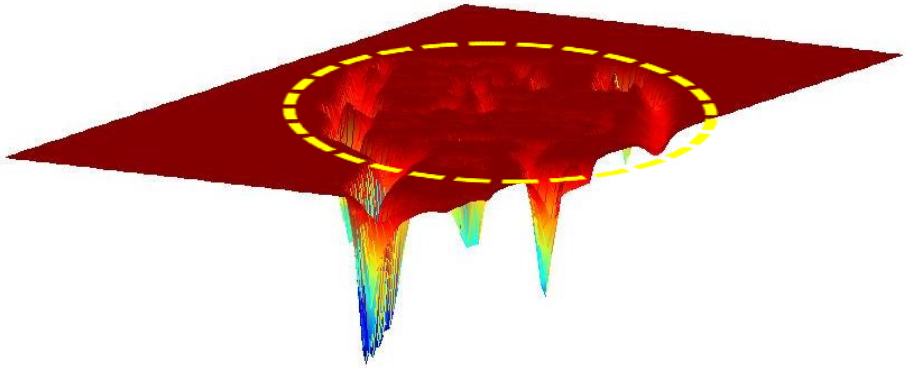
How do edges affect “snap” of rubber band?

Think of external energy from image as gravitational pull towards areas of high contrast



Magnitude of gradient

$$G_x(I)^2 + G_y(I)^2$$



- (Magnitude of gradient)

$$-\left(G_x(I)^2 + G_y(I)^2\right)$$

External image energy

- Image *I(x,y)*
- Gradient images $G_x(x,y)$ and $G_y(x,y)$

- External energy at a point *v(s)* on the curve is

$$E_{external}(v(s)) = -(\,|G_x(v(s))|^2 + |G_y(v(s))|^2\,)$$

- External energy for the whole curve:

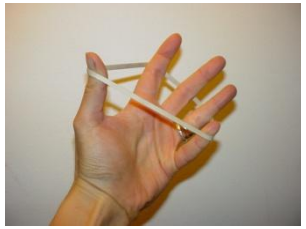
$$E_{external} = \int_0^1 E_{external}(v(s))\,ds$$

Internal energy: intuition

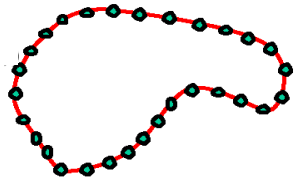
Internal energy: intuition

A priori, we want to favor smooth shapes, contours with low curvature, contours similar to a known shape, etc. to balance what is actually observed (i.e., in the gradient image).

http://www3.imperial.ac.uk/pls/portallive/docs/1/52679.JPG



Internal energy



For a continuous curve, a common internal energy term is the “bending energy”.

At some point *v(s)* on the curve, this is:

$$E_{internal}(v(s)) = \underbrace{\boxed{\alpha \left| \frac{dv}{ds} \right|^2}}_{\text{Elasticity, Tension}} + \underbrace{\boxed{\beta \left| \frac{d^2 v}{d^2 s} \right|^2}}_{\text{Stiffness, Curvature}}$$

The more the curve bends → the larger this energy value is.

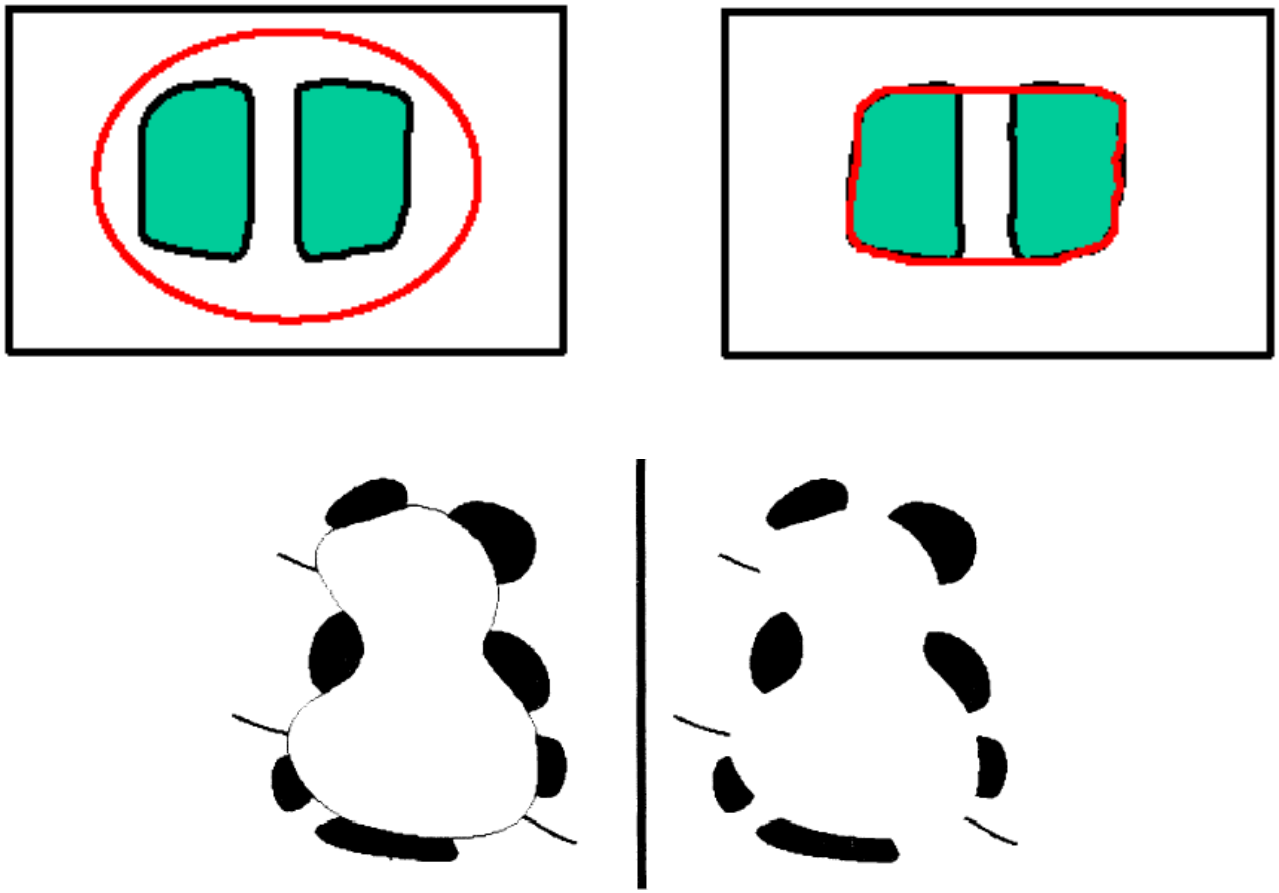
The weights α and β dictate how much influence each component has.

Internal energy for whole curve:

$$E_{internal} = \int_0^1 E_{internal}(v(s))\, ds$$

Dealing with missing data

- The smoothness constraint can deal with missing data:



[Figure from Kass et al. 1987]

Total energy
(continuous form)

$$E_{total} = E_{internal} + \gamma \quad E_{external}$$

$$E_{internal} = \int_0^1 E_{\text{internal}}(v(s))\,ds$$

// bending energy

$$E_{external} = \int_0^1 E_{external}(v(s))\,ds$$

// total edge strength under curve

Limitations

- External energy: snake does not really “see” object boundaries in the image unless it gets very close to it.

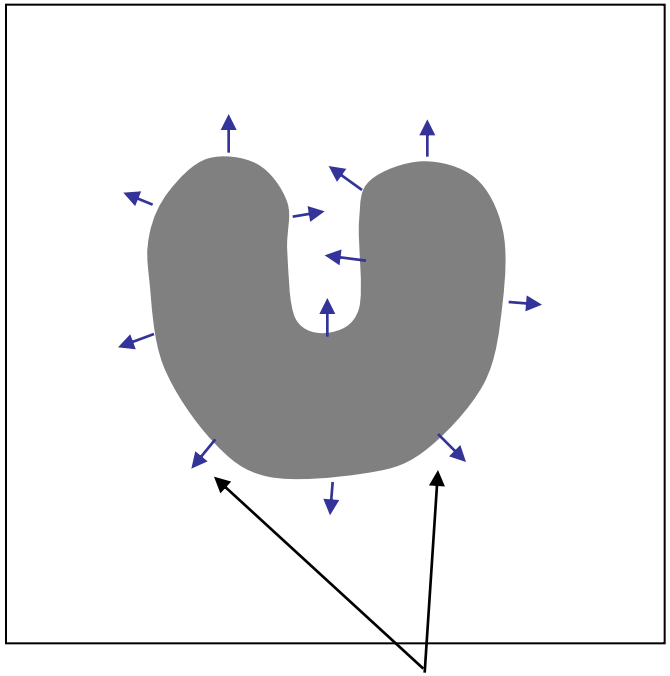
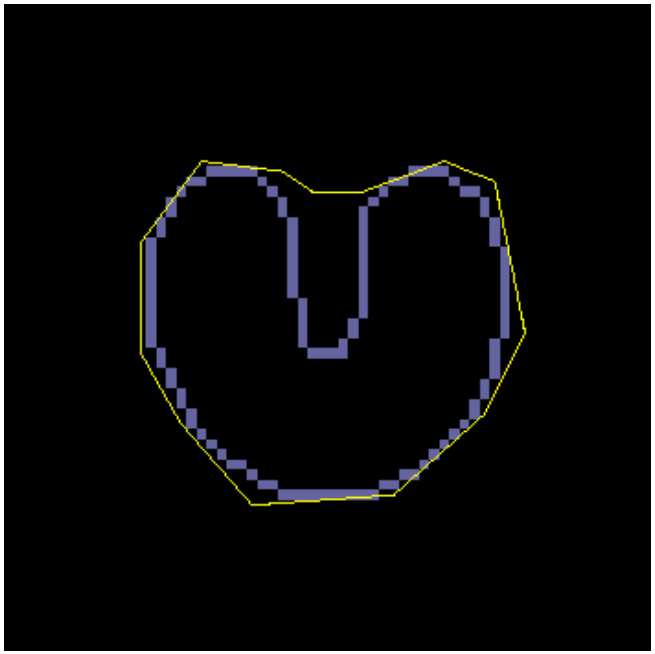



image gradients ∇I
are large only directly on the boundary